

PBPK / TOXICOKINETIC MODEL RESEARCH DOCUMENT

In vitro and computational approaches to predict developmental toxicity: Integrating PBPK models with cell-based assays

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SUMMARY & ABSTRACT:

Developmental toxicity assessment of Bisphenol A (BPA) and related chemical substitutes requires integrating in vitro cell-based assays with human physiologically based pharmacokinetic (PBPK) modeling. In this study, a multi-compartment human PBPK model was formulated to simulate oral administration, GI absorption, hepatic first-pass metabolism (glucuronidation to BPAG), systemic exposure, and tissue distribution to target organs during developmental windows.

ASSOCIATED MODEL & PARAMETER FILES (EXACT MATCHING NAME):

- PDF Document: Bisphenols_Moreau_2025.pdf
- Parameters CSV: Bisphenols_Moreau_2025_parameters.csv
- Antimony Model: Bisphenols_Moreau_2025_antimony.ant